



TATA TECHNOLOGIES INNOVENT-27

ChargeWise EdgeAI

Intelligent EV Charging & Battery Optimization at the Edge

Category: AI at the Edge Solutions for Automotive

Subcategory: Edge AI for Electric & Sustainable Mobility

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Charging apps find stations. They don't make decisions.



Discovery, not strategy

Existing apps locate the nearest charger but give no guidance on when, where, or how much to charge.



Battery health ignored

Habitual fast-charging to 100% accelerates degradation — the single costliest EV ownership expense.



Congestion at peak hours

Rising EV adoption concentrates demand at popular stations, extending queues during peak windows.



Range anxiety persists

Without route-aware planning, drivers over-charge defensively, wasting time and battery cycles.

India's EV inflection point strains charging intelligence

77%

YoY growth in electric car sales, 2025

FADA / electrive.com

29,000+

public charging stations installed nationwide (2025)

MoPNG

1 : 235

national EV-to-public-charger ratio vs. ~1:20 ideal

Industry estimate

Infrastructure is scaling faster than intelligence

Station count is growing roughly sixfold since 2022, but station-level software has not kept pace — utilization, queueing, and tariff-aware routing remain unsolved.

This gap is precisely where an edge-AI decision layer adds value: turning a growing but uneven network into a predictable charging experience.

What today's charging apps solve — and stop at

Category	Examples	Core function	Decision intelligence
OEM apps	Tata EV, MG, Hyundai, Ather	Vehicle telemetry, nearest charger	Minimal
Aggregators	PlugShare, Google Maps	Station discovery, reviews, navigation	None
Open data	OpenChargeMap	Crowd-sourced station database	None
Premium EV	Tesla	Route-aware Supercharger routing	Partial (cloud-based)

Every existing player answers “where can I charge?” — none answer “should I charge, and how much, right now?”

Where discovery-first tools fall short

-  No timing intelligence — electricity tariff and grid-load patterns are not factored into charge scheduling
-  No battery-health optimization — charge targets ignore degradation curves and thermal state
-  No wait-time prediction — station queues are shown as static occupancy, not forecast
-  Cloud dependency — latency and connectivity limit response in low-network zones
-  No route-aware charge planning — long trips are not decomposed into optimal charge stops

What ChargeWise EdgeAI is built to deliver



Cost-optimal timing

Recommend charge windows aligned to off-peak tariffs



Battery-safe targets

Suggest 80–90% charge levels to slow degradation



Smart station selection

Rank stations by distance, speed, and predicted wait



Route-aware planning

Decompose long trips into optimal charge stops



Edge-first inference

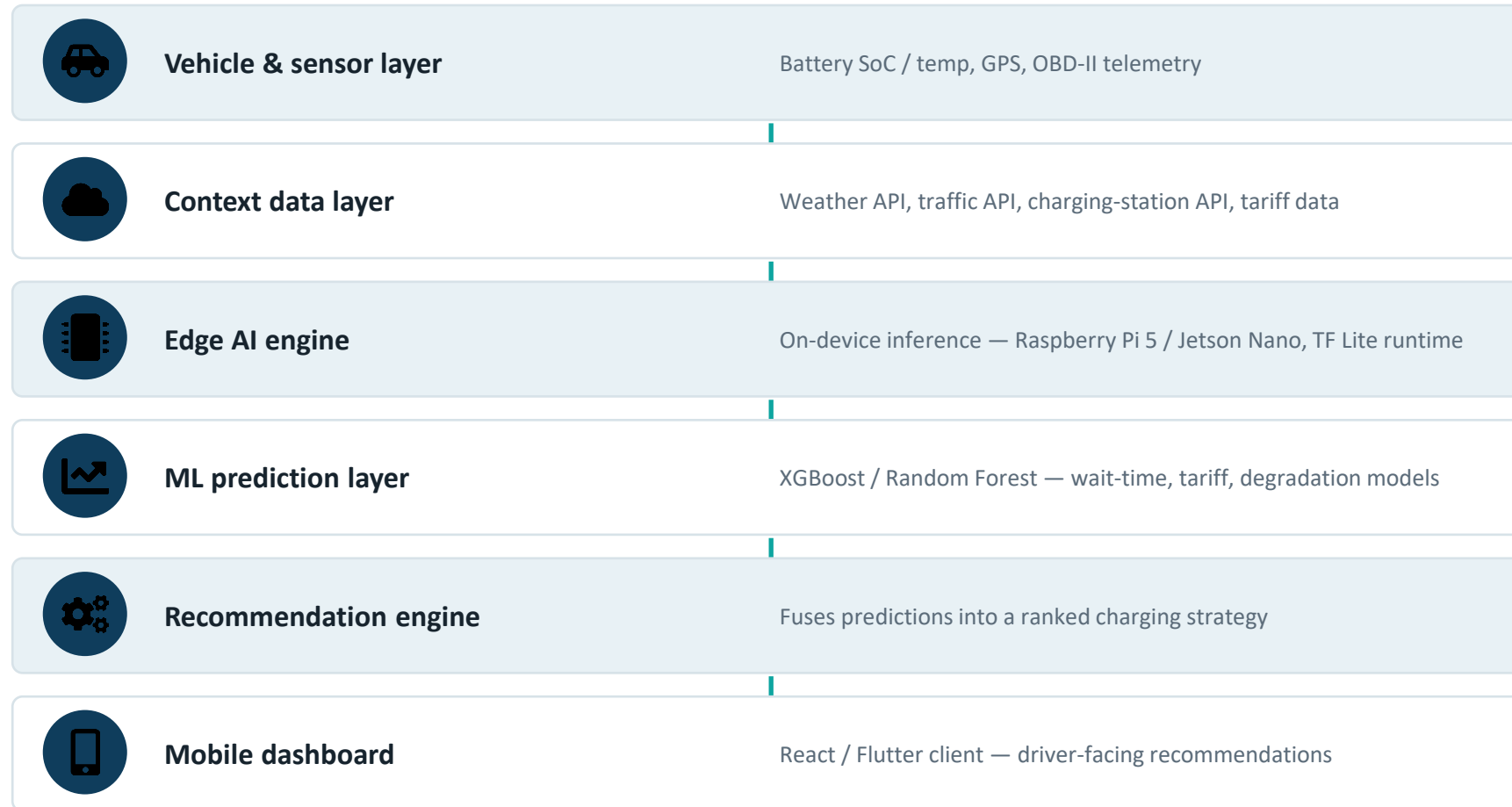
Run recommendation models on-device, sub-500ms



Privacy by design

Minimize cloud dependency for sensitive vehicle data

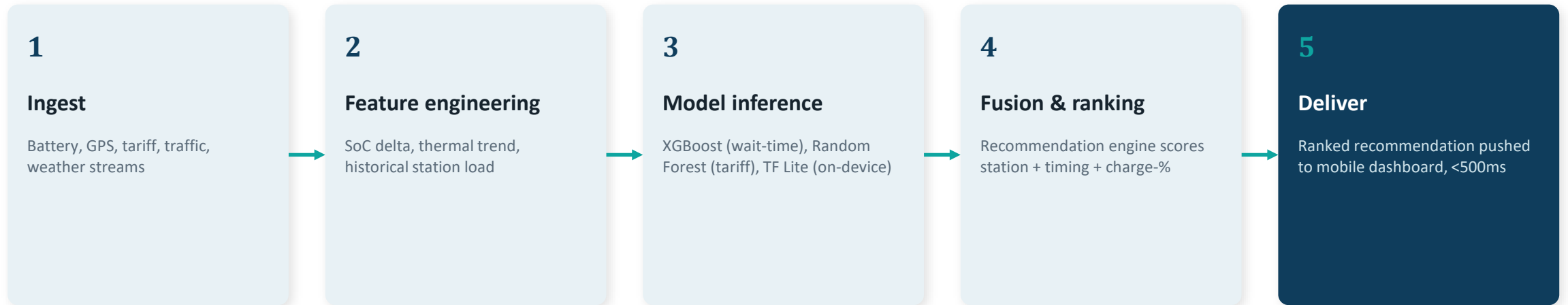
End-to-end architecture



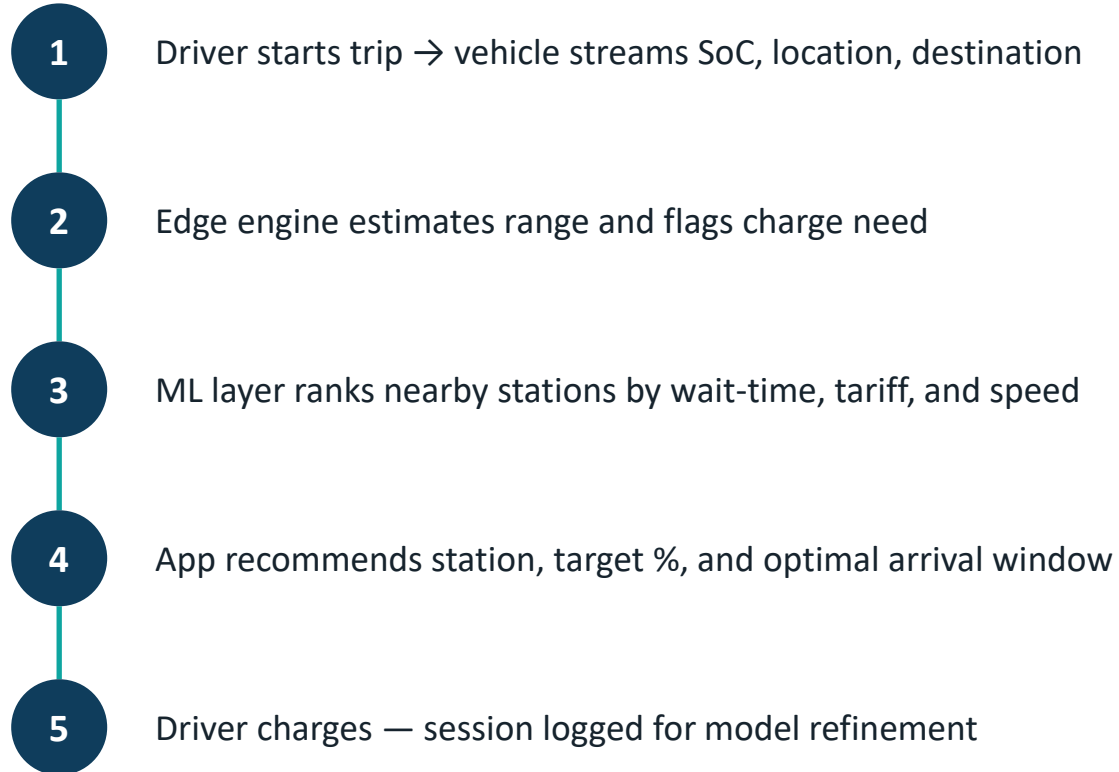
Design principle

Inference stays on-device. Cloud is used only for non-sensitive context data — weather, traffic, tariff feeds — never raw battery or location logs.

AI pipeline: from raw signal to recommendation



A single trip, decomposed into decisions



Stack by layer



AI / ML

Scikit-learn, XGBoost, Random Forest, TensorFlow Lite



Edge deployment

Raspberry Pi 5, Jetson Nano (prototype), TF Lite runtime, ONNX (optional)



Backend

Python, FastAPI, Flask



Frontend

React.js (web), Flutter (mobile)



Data & storage

SQLite, Firebase Firestore



Maps & data sources

Google Maps API, OpenStreetMap, OpenChargeMap API

Why inference happens on-device



< 500 ms

Local inference latency target for real-time recommendations



Privacy-preserving

Raw battery and location data never leave the vehicle



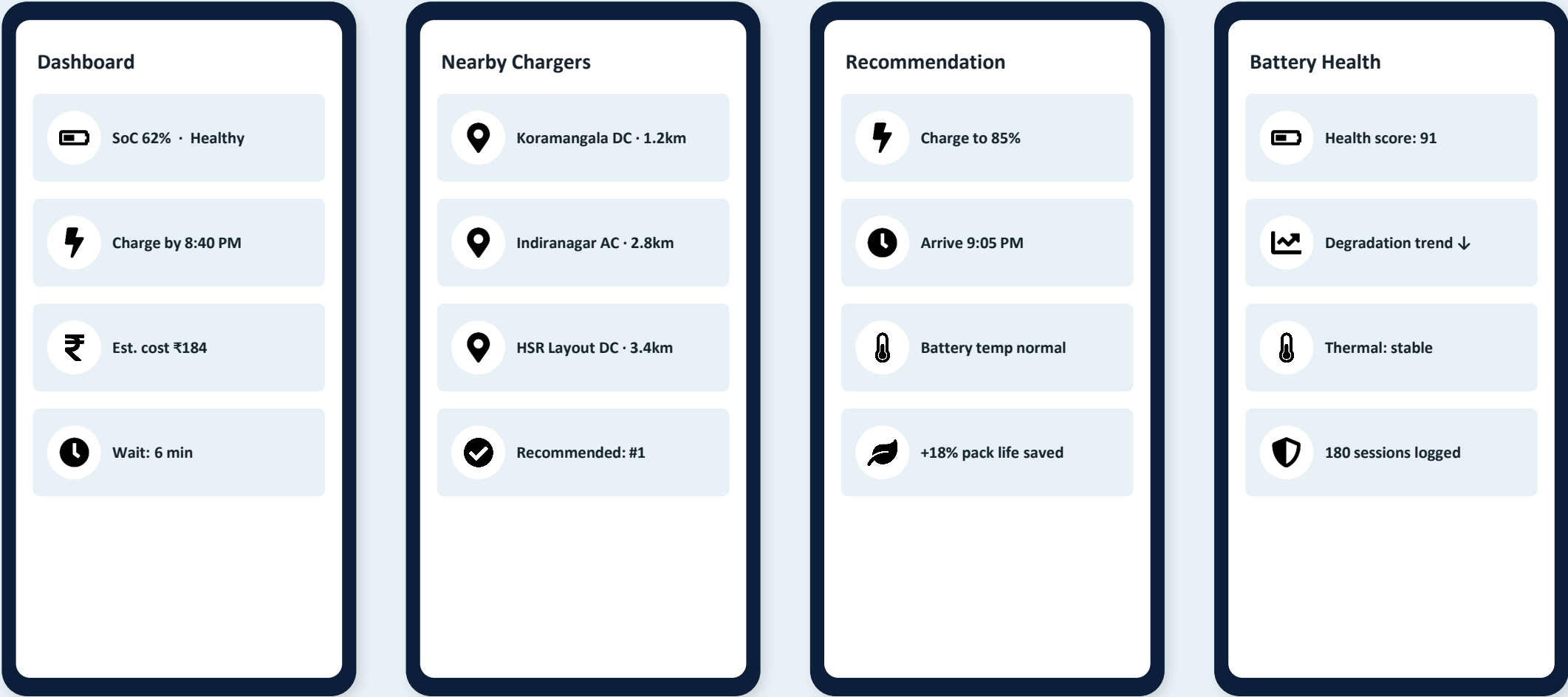
Works offline

Recommendations continue in low-connectivity zones

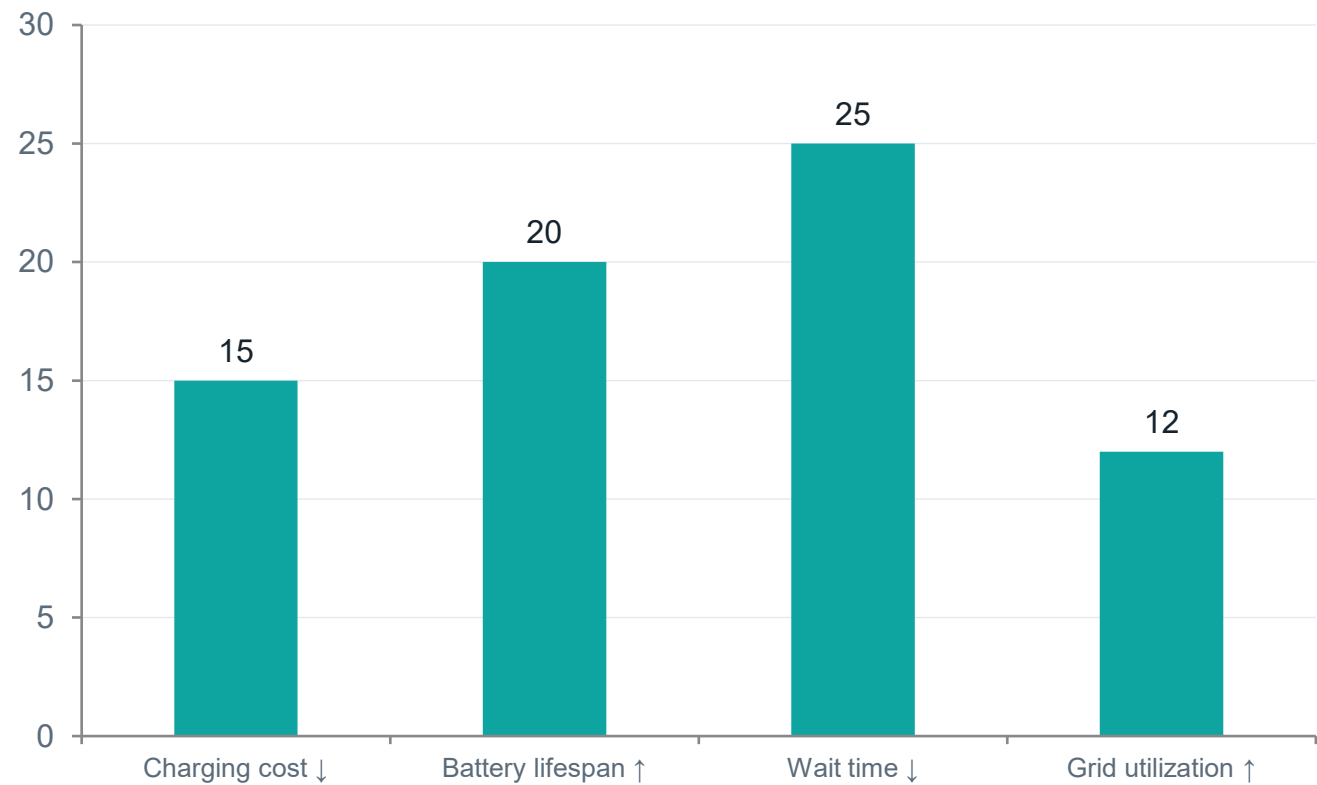
Hardware targets

- **Raspberry Pi 5**
Prototype control unit — sensor aggregation and inference host
- **NVIDIA Jetson Nano**
Evaluated for higher-throughput vision / sensor-fusion workloads

Mobile dashboard — concept



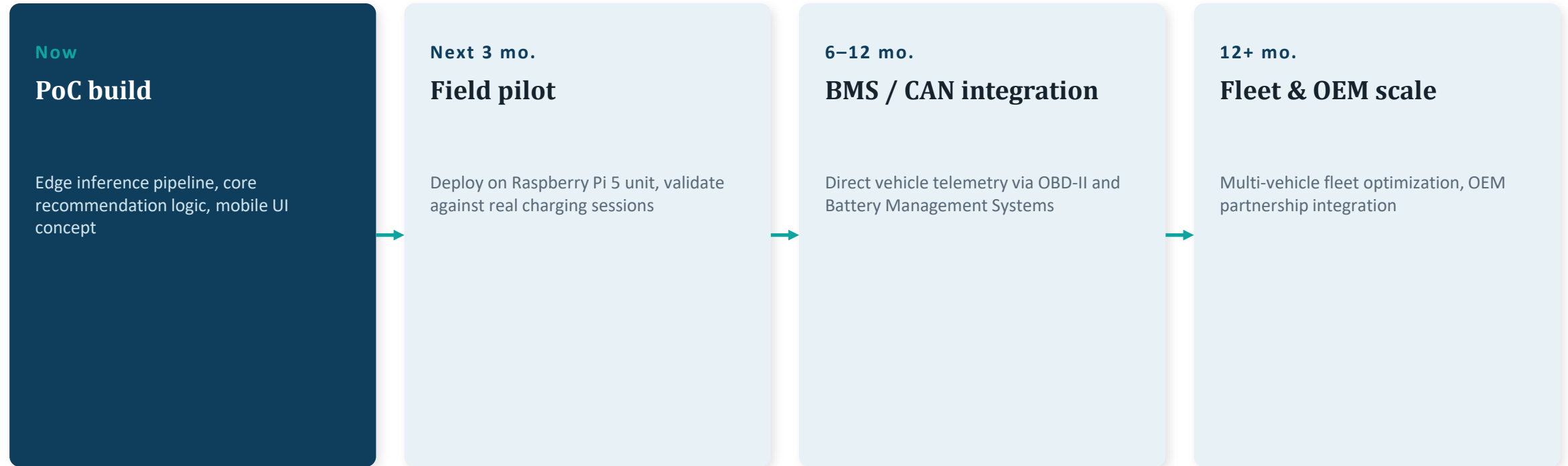
Projected outcomes from optimized charging



Midpoints of estimated ranges (see below) — not measured production results

Metric	Target range
Charging cost reduction	10–20%
Battery lifespan improvement	15–25%
Station wait-time reduction	20–30%
Edge inference latency	< 500 ms

Roadmap & future scope



Future scope

V2G-aware scheduling · solar-integrated charging bays · smart-grid demand response · degradation-aware second-life battery routing

Known risks, engineered around

Challenge	Mitigation
Limited access to real EV battery data	Public datasets + simulation-based validation
Dynamic charging station availability	Modular API integration layer
Edge hardware resource constraints	Lightweight TensorFlow Lite models
Real-time recommendation latency	Edge inference — no cloud round-trip

Every risk mapped to a specific engineering decision — not a hope.

Value created across the EV ecosystem



EV owners

Lower charging bills, healthier battery, less range anxiety



OEMs

Differentiated smart-charging feature, fewer warranty battery claims



Charging operators

Demand spread across the network, higher station utilization



Fleet operators

Lower per-km energy cost, predictable turnaround at depots



Government / policy

Efficient use of public charging capex, faster EV adoption



Grid operators

Demand-response ready, reduced peak-hour load spikes

Five advantages no competitor stacks together

-  Decision intelligence, not discovery — the only layer answering when, where, and how much to charge
-  Battery-health-aware by design — charge targets built around degradation curves, not just a full tank
-  Edge-first inference — sub-500ms recommendations, no cloud round-trip, works offline
-  Privacy-preserving — raw battery and location data stay on-device
-  Independently conceptualized — fresh architecture and ML pipeline, not a repackaged discovery app



Thank You

ChargeWise EdgeAI — making every charge a decision, not a guess.

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